

nAD10205-13g

10-bit 205 MSPS Analog-to-Digital Converter IP

FEATURES

- UMC L130E-HS 1.2 V Technology
- 10-bit Pipeline ADC
- 10 to 205 MSPS Conversion Rate,
205 MSPS with 8-bit accuracy,
165 MSPS with 9-bit accuracy
- Excellent Dynamic Performance
58 dBFS SNR at $F_{IN} = 10$ MHz
60 dBc SFDR at $F_{IN} = 10$ MHz
- Dynamic Power Scaling
~0.35 mW per MSPS
- Low Power Consumption
72 mW at 205 MSPS
28 mW at 80 MSPS
- Power Saving Idle Modes
- 0.48 mm² Core Area
- Pure Internal Voltage Reference
- Programmable Input Swing
0.5 to 1.0 V p-p Differential
- Excellent Gain and Phase Matching
- Input Clock Duty Cycle Restorer

APPLICATIONS

- Communication Receive Channel
WLAN 802.11x / WiMAX 802.16x
- Digital Imaging / Video
- TV / Video / Radio Decoders
- Graphic Capture

GENERAL DESCRIPTION

The nAD10205-13g is a monolithic, high-speed, ultra low power, dual channel analog-to-digital converter silicon IP. It uses a fully differential multistage pipeline architecture with digital error correction. The core includes a sample-and-hold and an internal voltage reference that provides a nominal full-scale range of 1.0 V peak to peak.

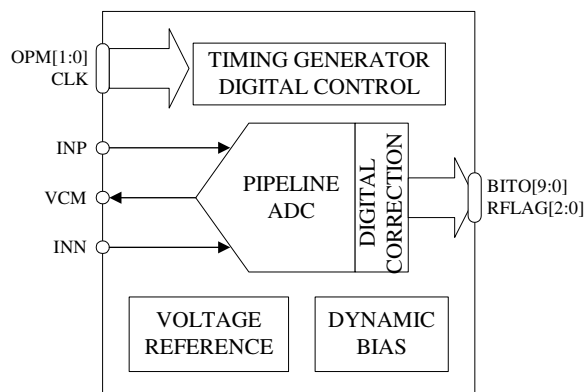


Figure 1. Functional block diagram

The ADC is designed for high dynamic performance at input frequencies up to Nyquist. It thus represents an ideal solution for demanding applications like broadband communication, TV/Video decoders and graphic capture.

The ADC consumes only 72 mW at 205 MSPS operation. Dynamic power scaling means that the power consumption scales linearly with approximately ~0.35 mW per MSPS. Combined with power saving idle modes the ADC is suitable for battery powered devices. Output data is available in a binary offset coded format. Three out-of-range indicator bits are also available for determining if the input signal is over-range, under-range or out-of-range.

Implemented in a generic 0.13 μ m CMOS process, employing a fully differential architecture and no required external reference decoupling it represents an ideal ADC for highly integrated mixed-signal systems.

QUICK REFERENCE DATA

IP Type / Technology	Hard Macro / UMC L130E-HS 1.2 V IP6M			
IP Area / Dimensions	0.48 mm ² / 1.2 × 0.4 mm			
Parameter	Min.	Typ.	Max.	Unit
Power Dissipation, at 205 MSPS		66		mW
Signal-to-Noise Ratio, at 205 MSPS $F_{IN} = 10$ MHz	54	58		dBFS
Spurious-Free-Dynamic Range, 205 MSPS $F_{IN} = 10$ MHz		65		dBc
Effective Number of bits, 205 MSPS $F_{IN} = 10$ MHz		7.7		Bit
Effective Number of bits, 165 MSPS $F_{IN} = 10$ MHz		9.3		Bit

Table 1. nAD10205-13g quick reference data

**ABOUT NORDIC SEMICONDUCTOR**

Founded in 1983, Nordic Semiconductor ASA (OSE:NOD) is a Fabless Semiconductor specializing in short-range radio communication and high-speed data conversion. Nordic is head quartered in Trondheim, Norway,

MIXED SIGNAL IP OFFERINGS

Leading and emerging Fabless Semiconductor companies around the globe leverage Nordic's portfolio of off-the-shelf mixed signal intellectual property for the design of highly integrated circuits for broadband communication, digital imaging and video.

Our IP portfolio includes medium resolution (8 to 12-bit), high-speed (40 MSPS+) data converters (ADC and DAC) and low jitter clock generation (PLL) implemented in the latest deep submicron CMOS technologies. Featuring low power consumption, low-silicon area, and a minimum of required external components and technology options our products offers cost efficient integration of high-performance data converters in cost sensitive, high volume integrated circuit designs.

DELIVERABLES

Our mixed signal intellectual property is delivered as technology specific hard macros. Upon licensing we provide a complete design-kit and documentation that enables efficient, low risk, design-in and integration. The design kit for our off-the-shelf IPs includes:

- Full datasheet
- Evaluation board and samples
- Silicon validation report
- Integration Guidelines
- Physical design database (flat gds2)
- LVS Netlist (spice compatible)
- Footprint (.lef format)
- HDL model (verilog model)
- Timing model (.lib)
- Design-in and integration support

FURTHER INFORMATION

For further information about this product, such as full datasheet, silicon validation report, availability and licensing terms please contact us at: datacon@nordicsemi.no

DOCUMENT INFO

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